Homework 6 Plots

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14 March 2022

3.6 Problems

Plot all solutions to the Heat Equation: $u_t = k * u_{xx}$. Evaluate the solutions and their behaviors. Discuss characteristics and similarities.

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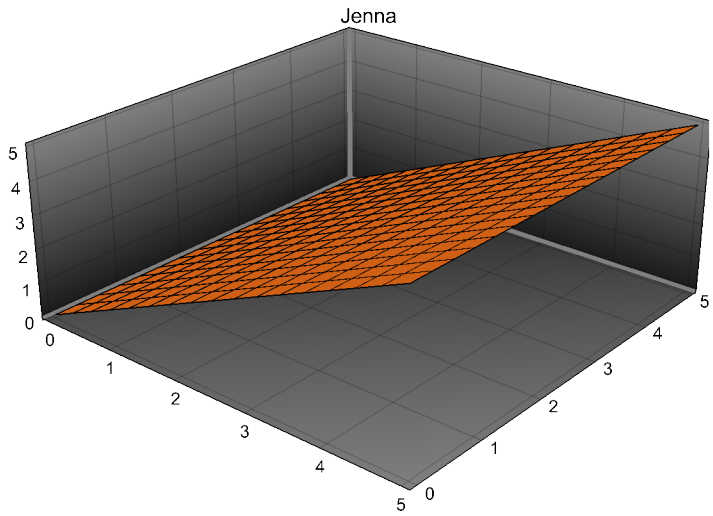
a.) u(0,t) = 0, u(1,t) = 1, f(x) = x, k = 2, and L = 1

For this problem, I obtained the general solution — u(x,t) = x — by solving for a particular linear equation and generalizing that equation for x and t. It's worth noting that the solution to this problem is identical to the problem presented in the notes. That said, this answer shouldn't be surprising since we generalized $u_p(x, t)$ in class and in the notes. It's also worth noting that this problems is identical to the example problem that was worked out in class.

In[21]:=
```
Jenna = u11[x_, t_] = x;
```

In[22]:=
```
Plot3D[Jenna, {x, 0, 5}, {t, 0, 5}, Mesh → Automatic, MeshStyle → Automatic,
 PlotTheme → "Marketing", PlotLegends → "Expressions", PlotLabel → "Jenna"]
```

Out[22]=



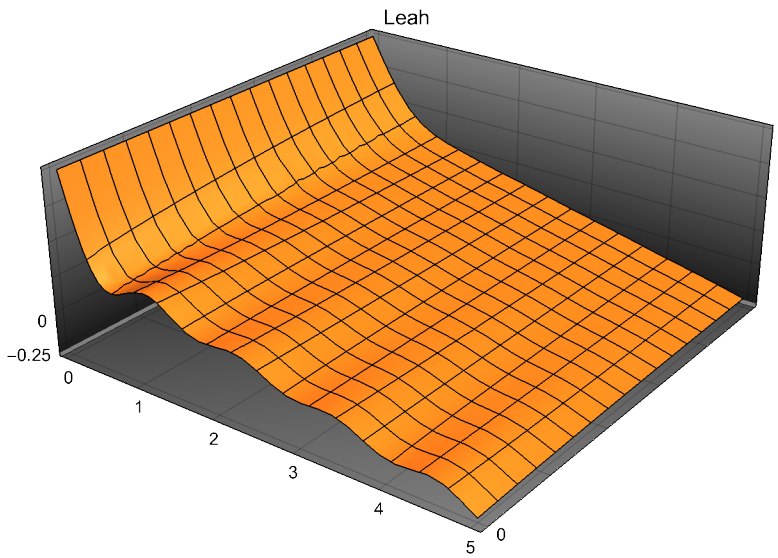
b.) u(0,t) = 1, u(π,t) = 0, f(x) = 1 - x, k = 1, and L = π

For this problem, I obtained the general solution — u(x,t) = $\frac{-x}{\pi} + 1 + \sum_{n=1}^{\infty} \frac{2-\pi}{\pi * n} * e^{-n^2 * t} * \text{Sin}[n * x]$ — by adding the particular linear equation to the fourier sine series. For simplicity sake, I will only evaluate a few terms of the series and plot the result here.

In[23]:=
```
Leah = u2[x_, t_] = -x/π + 1 + (2 - π)/π * e^(-t) * Sin[x] + ((2 - π) * e^(-4 t))/(2 * π) Sin[2 x] +
  ((2 - π) * e^(-9 t))/(3 * π) Sin[3 x] + ((2 - π) * e^(-16 t))/(4 * π) Sin[4 x] + ((2 - π) * e^(-25 t))/(5 * π) Sin[5 x];
```

In[24]:=
```
Plot3D[Leah, {x, 0, 5}, {t, 0, .1}, Mesh → Automatic,
 MeshStyle → Automatic, PlotTheme → "Marketing",
 PlotLegends → "Expressions", PlotLabel → "Leah"](*.1 originially *)
```

Out[24]=



Here, we can observe what happens when we have both a homogeneous and particular solution since the above graph has the downward slope of the particular solution and the sinusoidal behavior of the fourier series. In short, it looks like what we would expect the product of two functions would look like.

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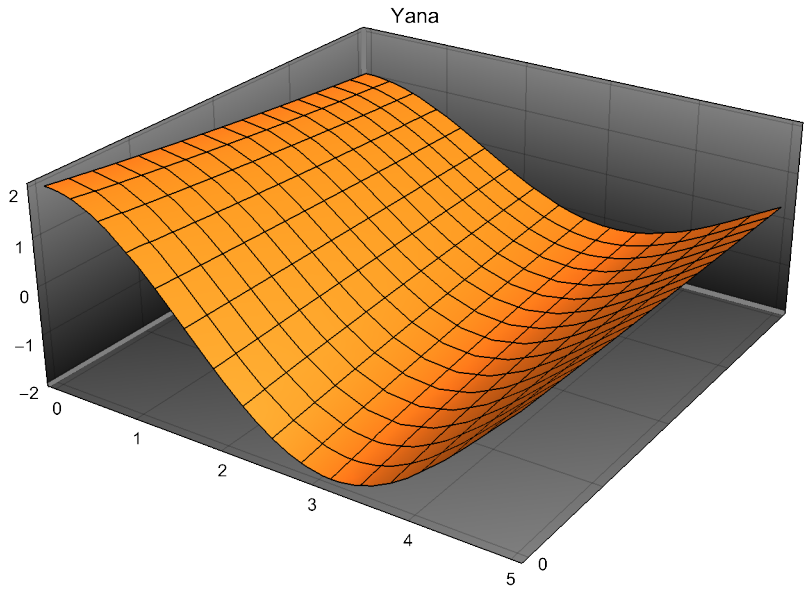
c.) u(0,t) = 0, $u_x(\pi,t) = 0$, f(x) = Sin[x], k = 3, and L = π These are the correct, updated conditions that we discussed in Office Hours.

First, it's important to point out that the original problem had a typo in that $u_x(\pi, t) = 0$ is the correct initial condition while $u_x(\pi, t) = -1$ was initially given. That aside, this problem exhibits Homogeneous Neumann Conditions and can be solved as such; our final solution being $u(x, t) = 2\,e^{-3t} * \text{Cos}[x]$.

In[25]:=
```
Yana = u3[x_, t_] = 2 * e^(-3 t) * Cos[x];
```

In[26]:=
```
Plot3D[Yana, {x, 0, 5}, {t, 0, .2}, Mesh → Automatic, MeshStyle → Automatic,
 PlotTheme → "Marketing", PlotLegends → "Expressions", PlotLabel → "Yana"]
```

Out[26]=



In the above graph, we can clearly see the exponential decay and also briefly see the beginning of the sinusoidal behavior as the graph begins to go back up as x increases. It's also worth noting that both b and c exhibit visual sinusoidal behavior since both solutions are partially composed of sinusoidal terms.

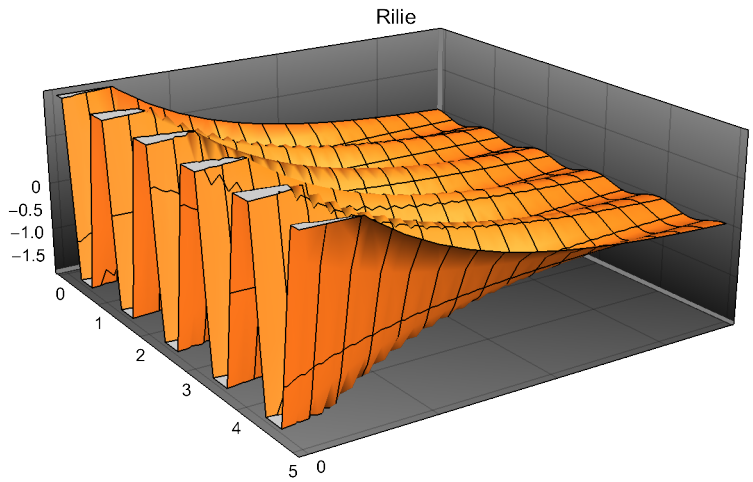
d.) $u_x(0,t) = 0$, $u_x(2,t) = 0$, f(x) = 3Cos[2πt], k = $\frac{1}{2}$, and L = 2

For this problem, I obtained the general solution — $u(x, t) = 3\,\mathrm{Cos}[2\,\pi * x] * e^{-2\,\pi^2 * t}$ — by applying the technique for Homogenous Neumann Conditions. Since this solution is not represented by a series, plotting this function should be a little easier.

In[27]:=
```
Rilie = u4[x_, t_] = 3 Cos[2 π * x] * e^(-2*π^2*t);
```

In[28]:=
```
Plot3D[Rilie, {x, 0, 5}, {t, 0, .18}, Mesh → Automatic,
 MeshStyle → Automatic, PlotTheme → "Marketing", PlotLabel → "Rilie"]
```

Out[28]=



In the above graph, we can clearly see both the sinusoidal and exponential behavior of the solution. However, since the exponential behavior is so much more dominant, we can see the exponential behavior dominate towards the front of the graph, and taper off to a complete sinusoid towards the back.

3.) Discuss the variation in solutions from the last homework with homogenous Dirichlet boundary conditions. How did the change in conditions affect the solutions?

In our previous homework, the homogeneous Dirichlet boundary conditions lead the heat distribution in our rod to go to zero without variation. While this was convenient from a computational standpoint, it severely limits the use of such equations in applied settings. Luckily, we are now exploring some more complicated cases! In parts a. and b. we still deal with Dirichlet boundary conditions, but the left and right sides have been set to one respectively. In application, this means the heat distribution will not ever go to zero. This makes sense, since the problem is defined such that one side is zero and one side is 1 for all values of t greater than zero . Interestingly, while looking at the graphs you may note that they are tilted in opposite directions along the x-axis. This is consistent with the fact that the non-zero values are on opposite ends for these two problems.

The 2D heat equation is nice to work with since their is only one constant you have to solve for, which is Bm,n unlike in the wave equation where you have to solve for Am,n as well as Bm,n. The 2D wave equation also involves the sine series.

Part d has us dealing with homogeneous Neumann boundary conditions. Since they are defining the rate of temperature change at the boundaries as zero, this would be the case where the ends of the rod are insulated. This has the neat effect of smoothing the heat distribution over time, but at a much slower rate than we have seen in previous problems. In this solution we still see the heat going to zero along the length of the rod, and given enough time, it would flatten out completely. In this case however, this is not due to the boundaries being set to zero , but that the integral of the heat distribution along the x-axis at t=0 is equal to zero. If we were dealing with an f(x) which were not so symmetrical, we would likely find the heat distribution tending toward some other constant.

Neumann boundary conditions define the rate of change at the edge of a bounded curve in 2D. These conditions will always result in a cosine series. These conditions are often called “insulating boundaries” since they occur at boundaries and maintain them. In the heat equation this represents how the temperature is changing at boundary.

With the benefit of hindsight, we can now compare our current solutions to the solutions of the previous homework. The results: well, they’re a bit disappointing. In both of these two homework assignments, we often get identical solutions to certain problems. I mean, in both this week and last we got solutions that just ended up being planes in space with other solutions resembling a sheet of paper. But, I guess I can’t be too upset since the overwhelming amount of PDEs in the world are either unsolvable or have no real meaning. Maybe an advanced course on the topic would cover other, unique solutions; but I guess I should be content with what I have right now.

In terms of differences, the most evident example is the pronounced sinusoidal behavior in this homework assignment. Problem “d” in-particular exhibits some pretty extreme sinusoidal behavior that’s a direct result of the large exponential term $-2\,\pi^2$. I also think that the reason *why* these problems exhibited more “extreme” behavior in general is that we’re solving problems with more variables and, as such, more things need to be accounted for.

3.7 Problems

Plot all solutions to the 2D Wave Equation: $u_{tt} = c^2(u_{xx} + u_{yy})$. Evaluate the solutions and their behaviors. Discuss characteristics and similarities.

a.) u(x,y,0) = Sin[2x]*Sin[y], $u_t(x, y, 0) = 0$, c = 2, a = b = π

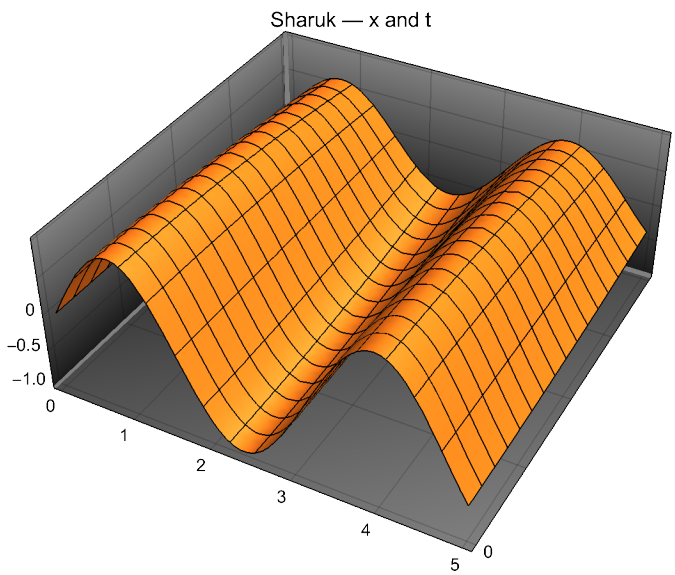
For this first 2D solution to the wave equation, I obtained a pretty straightforward solution — u(x,y,t) = Sin[2x] * Sin[y] * Cos$\left[2\sqrt{5} * t\right]$ — given the above criteria. Much like the first problem to 3.6, this problem is identical to the in-class counterpart and was not terribly difficult to solve. Now let's take a look at a physical representation of the function.

In[29]:=
```
Corrine = u5[x_, y_, t_] = Sin[2 x] * Sin[y] * Cos[2 √5 * t];
Sharuk = u51[x_, t_] = Sin[2 x] * Cos[2 √5 * t];
Ian = u52[y_, t_] = Sin[y] * Cos[2 √5 * t];
```

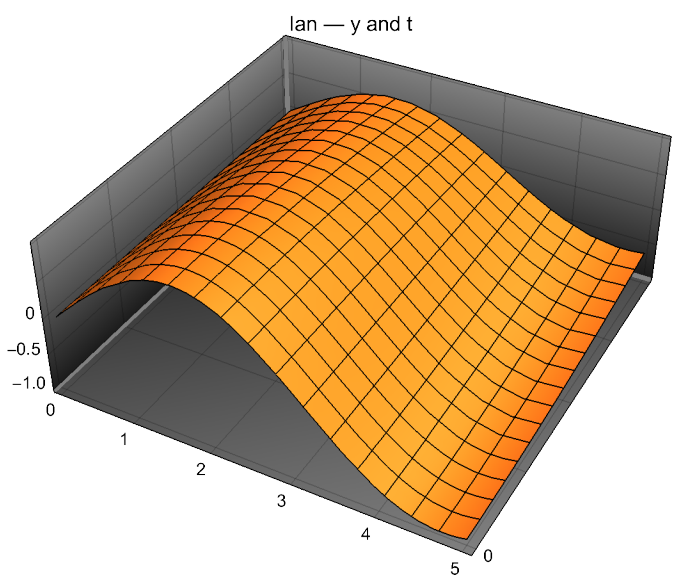
In[32]:=
```
Plot3D[Sharuk, {x, 0, 5}, {t, 0, .2}, Mesh → Automatic,
 MeshStyle → Automatic, PlotTheme → "Marketing",
 PlotLegends → "Expressions", PlotLabel → "Sharuk - x and t"]

Plot3D[Ian, {y, 0, 5}, {t, 0, .2}, Mesh → Automatic, MeshStyle → Automatic,
 PlotTheme → "Marketing", PlotLegends → "Expressions", PlotLabel → "Ian - y and t"]
```

Out[32]=



Out[33]=



In both of the above examples, we can clearly see the sinusoidal behavior that we were expecting from our solution, and also enough of a difference to be reflective of the additional variable.

b.) u(x,y,0) = Sin[π*x]*Sin[y], $u_t(x, y, 0)$ = Sin[$\pi * x$], c = 1, a =

1, b = π

To solve this particular equation, I needed to use both previous techniques for multiple variables as well as techniques for coefficients of multiple variables. Additional tactics aside, this wasn't too bad despite the additional computation.

In[35]:=
```
rick = u6[x_, y_, t_] = Sin[π * x] * Sin[y] * (Cos[Sqrt[π^2 + 1] * t] + Sin[Sqrt[π^2 + 1] t] / Sqrt[π^2 + 1]);
Hailey = u61[x_, t_] = Sin[π * x] * (Cos[Sqrt[π^2 + 1] * t] + Sin[Sqrt[π^2 + 1] t] / Sqrt[π^2 + 1]);
Doug = u62[y_, t_] = Sin[y] * (Cos[Sqrt[π^2 + 1] * t] + Sin[Sqrt[π^2 + 1] t] / Sqrt[π^2 + 1]);
```

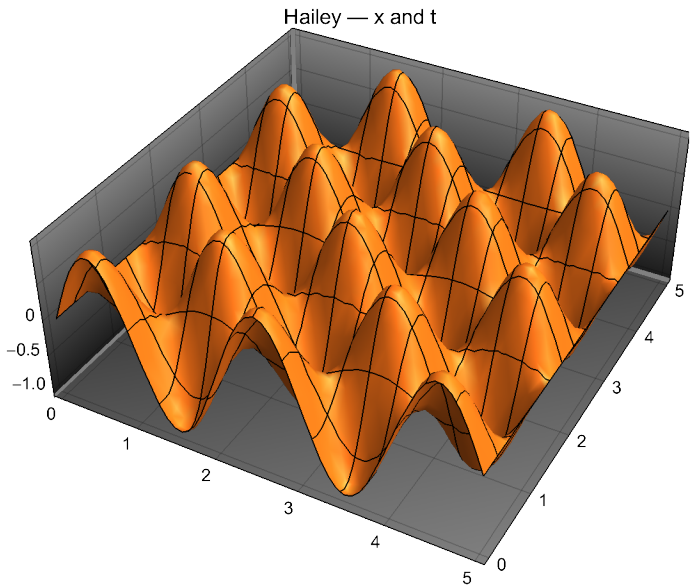
$$\text{rick} = u_6[x_, y_, t_] = \text{Sin}[\pi * x] * \text{Sin}[y] * \left(\text{Cos}\left[\sqrt{\pi^2 + 1} * t\right] + \frac{\text{Sin}\left[\sqrt{\pi^2 + 1}\ t\right]}{\sqrt{\pi^2 + 1}}\right);$$

$$\text{Hailey} = u_{61}[x_, t_] = \text{Sin}[\pi * x] * \left(\text{Cos}\left[\sqrt{\pi^2 + 1} * t\right] + \frac{\text{Sin}\left[\sqrt{\pi^2 + 1}\ t\right]}{\sqrt{\pi^2 + 1}}\right);$$

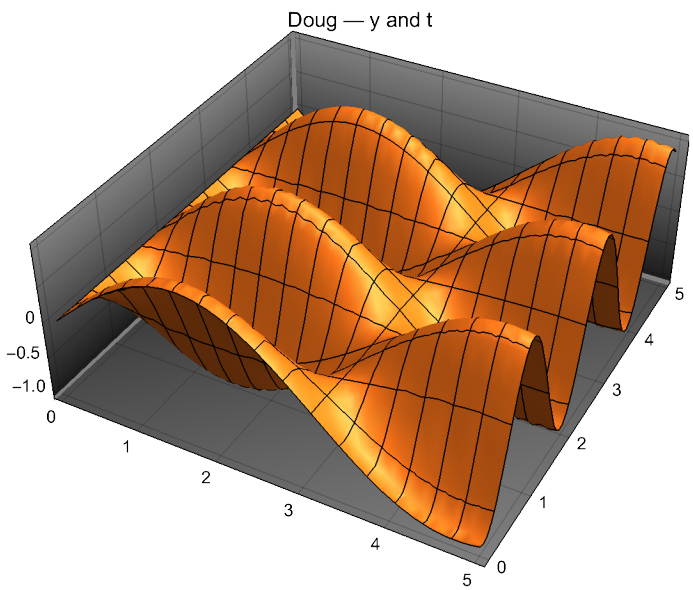
$$\text{Doug} = u_{62}[y_, t_] = \text{Sin}[y] * \left(\text{Cos}\left[\sqrt{\pi^2 + 1} * t\right] + \frac{\text{Sin}\left[\sqrt{\pi^2 + 1}\ t\right]}{\sqrt{\pi^2 + 1}}\right);$$

In[38]:=
```
Plot3D[Hailey, {x, 0, 5}, {t, 0, 5}, Mesh → Automatic,
 MeshStyle → Automatic, PlotTheme → "Marketing",
 PlotLegends → "Expressions", PlotLabel → "Hailey - x and t"]
Plot3D[Doug, {y, 0, 5}, {t, 0, 5}, Mesh → Automatic, MeshStyle → Automatic,
 PlotTheme → "Marketing", PlotLegends → "Expressions", PlotLabel → "Doug - y and t"]
```

Out[38]=



Out[39]=



Again, we get two sinusoidal pictures that are consistent with our solution. However, this example has two solutions that are even more unique than in part a.

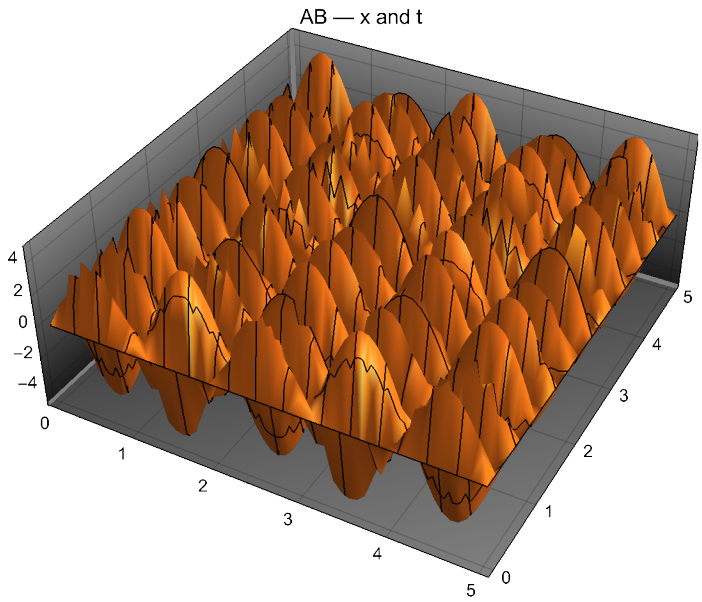
c.) u(x,y,0) = 0, $u_t(x, y, 0) = 1$, c = 3, a = b = 1

I couldn't think of anything to say here, so let's just get straight to the graphs.

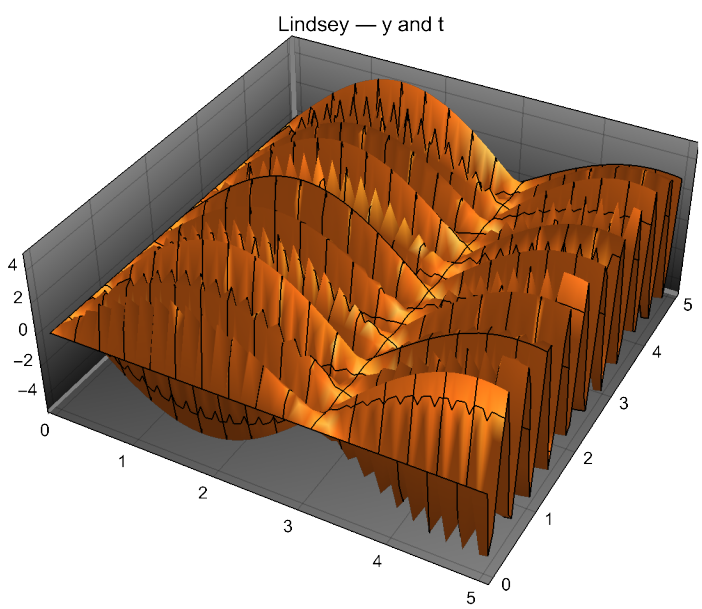
In[40]:=
```
Chad = u7[x_, y_, t_] = (48/π^2) * Sin[π * x] * Sin[y] * Sin[3 * π * √2 t];
AB = u71[x_, t_] = (48/π^2) Sin[π * x] * Sin[3 * π * √2 t];
Lindsey = u72[y_, t_] = (48/π^2) Sin[y] * Sin[3 * π * √2 t];
```

In[43]:=
```
Plot3D[AB, {x, 0, 5}, {t, 0, 5}, Mesh → Automatic, MeshStyle → Automatic,
 PlotTheme → "Marketing", PlotLegends → "Expressions", PlotLabel → "AB - x and t"]
Plot3D[Lindsey, {y, 0, 5}, {t, 0, 5}, Mesh → Automatic,
 MeshStyle → Automatic, PlotTheme → "Marketing",
 PlotLegends → "Expressions", PlotLabel → "Lindsey - y and t"]
```

Out[43]=



Out[44]=



This solution is pretty similar to that of part b. However, these pictures have some “spikes” protruding

from the pictures with that make them a little harder to view conceptually, but are still valid solutions.

3.) What is interesting, or of note about the solutions in 2D?

Explain how the additional spatial variable affected the solution form.
How would you expect such solutions to appear for the 2D Heat Equation?
Extrapolate and explain what your expectations are for problems that have nonhomogeneous boundary conditions.
Then discuss the effects of conditions such as homogeneous Neumann BCs for 2D problems.

For these 2d problems, I needed to set my solutions in terms of three variables: x, y and t. Aside from making the problems all the more difficult, I also needed to deal with only a few solutions terms at a time. If a way to graph these three variable functions exists in Mathematica, I couldn't find it. However, I do know that an animation function exists in the program, but I could not figure out how to use it given the assignment's time constraints. Problems aside, let's dive into the analysis.

Given the additional spatial variable, I decided the best way to graph the 2d problems would be to break them up; and so I did. In doing so, I discovered that the solutions tend to follow a similar pattern in that the solutions tend to be waves themselves and that they all propagate in the same general direction. If I could figure out how animate works, I might be able to see this propagation in action, but I've already said what I want about that.

If I were to find particular 2d solutions to the heat equation, I'd expect similar problems and would have to break up any and all solutions that I obtained. Problems aside, I also think that solutions would follow a similar pattern to what we experienced here in that the solutions to unique problems would share a physical resemblance to each other as well as the solutions of the individual.

When given any 2d P.D. wave equation, I would expect any and all solutions to resemble physical

waves, go figure. Sarcasm aside, this has an intuitive level of sense to it since all waves have a peak, trough, wavelength, amplitude and other characteristics that one would typically associate with waves. As for the boundary conditions, I think the most reasonable expectation to hold would be that our final solution may or may not experience damping from any forcing function. This is also consistent with what was previously discussed in engineering/ scientific applications to P.D.E.s as well as our own personal experience. Natural waves—be they ocean waves, waves on a string, etc.—all tend to come to a stop since they experience external forces that cause their motion to end. However, given modern technology, we can experience what pure, unending waves are like and describe their motion.

Nonhomogeneous boundary conditions require us to find a particular Yp first with those nonhomogenous conditions. The solution to this will theoretically solve the whole PDE by itself. From here we would find a homogenous Yh with homogenous their homogeneous conditions and add it to the Yp. The addition of Yp + Yh will solve the whole, actual, PDE and not the theoretical one that results from Yp alone. Nonhomogenous side conditions occur when the conditions exceed the range of our current methods to solve PDEs and thus require us to change into an equivalent form that are solvable using S.O.V.. then add to original solution, again, this is the Yp.

Dirichlet: Defined on the function boundary itself.
-With the above definition, we can infer that Dirichlet conditions will only *directly* impact the boundaries, but the rest of the function will be impacted as a result.

Neumann: Defined on the derivatives.
-Since Neumann conditions directly impact the derivatives, we need to take this into account and also realize that we’re working with a rate of change and take antiderivatives when appropriate.

Robin: Defined on a linear combination of the two.
-For Robin conditions, we’re going to have to take both of the above observations into account, apply dual methods of solving the equations, and also account for anything else that may come from the problem.

It’s also worth noting that, since we’re solving these PDEs using SOV, our solutions will always be a product of however many variables our problem uses.

In conclusion, this homework assignment was a lot of work and produced some interesting results. Only time will tell when this will come up again, but for now, I think that we have pretty good understanding of the concepts covered in this assignment and are looking forward to what comes next.